

EDITORIALS

Obesity and CKD: How to Assess the Risk?

For many years, the adverse effect of obesity on kidney outcomes has been recognized in patients with primary kidney diseases in general¹ as well as in patients with specific diseases such as immunoglobulin A glomerulonephritis.² There is also evidence that reversal of obesity improves albuminuria^{3,4} and glomerular hyperfiltration in patients with morbid obesity.⁵ Even in the absence of primary kidney diseases, obesity may cause increased glomerular size and glomerular function abnormalities, may also cause a unique form of focal segmental glomerulosclerosis (FSGS) with severe proteinuria, and is often accompanied by rapid loss of kidney function.^{6,7}

However, the problem is more far-reaching. In the Western world, obesity has recently become an ever more important risk factor for development of chronic kidney disease (CKD), manifesting with low glomerular filtration rate (GFR) or proteinuria. Two long-term observational studies documented a correlation between body mass index (BMI) and subsequent onset of end-stage renal disease (ESRD) in Japanese men⁸ and US Americans.⁹ Interestingly the threshold BMI values for the risk of ESRD were different between Asian and non-Asian populations^{8,9}; while in white Americans, the risk increased significantly beyond a BMI of 25 kg/m², the threshold in Japanese men was 21 kg/m²—in line with the finding of a greater proportion of visceral fat and higher cardiovascular risk at any level of BMI in Asians compared to whites. As a consequence, in Asians, a lower threshold value of BMI for defining obesity has been recommended for preventive measures.¹⁰

It is not only obesity at adult age, but even overweight in adolescents, which is predictive of CKD in adult age: Ejnerblad¹¹ showed that BMI

at age 20 years is an important determinant of CKD in adult life. Compared to a BMI less than 25 kg/m², the odds ratio for advanced CKD (serum creatinine > 3.4 mg/dL), admittedly without adjusting for diabetes, was 3.1 (in men) and 3.0 (in women), respectively, for a BMI greater than 25 kg/m². This consequence of body weight at young age is of interest, since it may point to one (certainly not the only) link between obesity and CKD: intrauterine stress. Poor in utero conditions may, by fetal programming (according to Barker's hypothesis), lead to development of fewer nephrons and therefore predisposition to CKD and hypertension in adult life,¹² but in parallel also to obesity, metabolic syndrome, and high cardiovascular risk.¹³

Against this background, it is not a surprise that in this issue of the *American Journal of Kidney Diseases*, 5 original research articles address the topic of obesity and CKD, an issue of substantial public health relevance.

As part of the Framingham Heart Study program, Foster et al¹⁴ prospectively followed a cohort of 2,676 participants in the Framingham Offspring Study (roughly two-thirds of the original cohort) over a 18.5-year period. They assessed the association between BMI and onset of CKD stage 3, defined using sex-adjusted estimated GFR (eGFR) thresholds of greater than 59 mL/min/1.73 m² in women and 64 mL/min/1.73 m² in men, as has been suggested by Poggio and

Address correspondence to Eberhard Ritz, MD, Nierenzentrum, Im Neuenheimer Feld 162, Heidelberg, Germany D 60100. E-mail: prof.e.ritz@t-online.de

© 2008 by the National Kidney Foundation, Inc.

0272-6386/08/5201-0001\$34.00/0

doi:10.1053/j.ajkd.2008.05.002

Rule.¹⁵ In this cohort, which is sadly representative for Western societies, 36% were overweight and 12% outright obese. A total of 7.9% developed CKD stage 3 and—equally important—14.4% developed proteinuria. Obese individuals had a 68% increased odds of developing CKD stage 3, but such a pronounced relation between abnormal BMI and CKD was no longer statistically significant after adjustments were introduced in multivariable models.

The absence of statistical significance does not disprove the relationship between obesity and CKD. Although the study is limited to some extent by its restricted size, this is compensated by the long duration of observation and the high quality of study performance and statistical analysis. It is increasingly recognized that not all body fat is created equal: visceral fat confers a substantially greater cardiovascular¹⁶ and renal risk¹⁷ than does subcutaneous fat, as beautifully illustrated by the observation that liposuction, ie, selective removal of subcutaneous fat while leaving untouched visceral fat, fails to improve insulin resistance, a strong marker of cardiovascular risk, as well as other facets of the metabolic syndrome.¹⁸ Possibly a stronger relationship would have been found to visceral fat, assessed for instance by the simple maneuver of measuring the waist circumference or waist-to-hip ratio as illustrated by the paper of Elsayed et al,¹⁹ also published in this issue.

The interpretation of the data of this study raises another less obvious issue as well. For a long time, opinions have been divided whether GFR values should be adjusted for body surface area (BSA) or not. This issue is particularly relevant for the assessment of kidney function in obese individuals. Even Homer W. Smith, who pioneered the development renal clearance measurement methods, admitted that adjustment to a standard BSA of 1.73 m² is completely arbitrary.²⁰ The value of 1.73 m² is based on measurements in fewer than 10 white individuals, rendering generalization somewhat problematic. Accordingly, at any given level of GFR, lower values of GFR must result when GFR is divided by the (higher) BSA of obese individuals. The Solomonic solution of this issue would be to analyse and report both the adjusted and unadjusted values. One may even argue that the “workload” (however defined) imposed upon

the glomerulus, which is responsible for glomerular damage, is less well captured by the GFR than by the glomerular pressure. In humans, this parameter cannot be directly measured in vivo, but to some extent it is reflected by the filtration fraction.

Another methodological issue is the categorization of individuals as overweight or obese. Bosma et al documented that a continuous positive relationship exists between BMI and increased filtration fraction throughout the entire range from normal to abnormal BMI values.²¹ In this context, it is of interest that, even in men as young as 18 years old, unadjusted estimated creatinine clearance is progressively higher in individuals with progressively higher BMI.²² Such hyperfiltration is one aspect which bedevils assessment of GFR loss. It is reasonable to assume that obesity-induced single-nephron hyperfiltration persists even when a substantial number of glomeruli have been lost, so that the whole kidney GFR may well underestimate the magnitude of nephron loss. The causes underlying hyperfiltration in obesity are not fully clarified. It is unlikely that a single cause is responsible. It has been shown that excess sodium intake increases GFR more in obese individuals,²³ possibly via extracellular volume expansion or via secretion of cardiogenic steroids.²⁴ In animal experiments, abnormal tubular feedback with increased sodium reabsorption in the loop of Henle caused preglomerular vasodilatation and raised glomerular pressure.²⁵ Insulin resistance is also related to increased GFR and filtration fraction,²⁶ although this is presumably not mediated directly by higher insulin levels, as insulin infusion does not reproduce the effect.²⁷ A contributory factor may be glomerulomegaly, resulting in a high filtration surface.²⁸

Decreased GFR is not the only manifestation of CKD that may be due to obesity. An important observation of the study by Foster et al in this issue is the finding that, in overweight and obese patients, the risk of incident dipstick proteinuria is 43% to 56% higher than in individuals with BMI below 25 kg/m².¹⁴ Why is proteinuria so important in this context? For risk assessment in CKD, it is important to distinguish between individuals without proteinuria (for instance, those with age-associated reduction in eGFR who are at low risk for kidney failure and cardio-

vascular events) and individuals with proteinuria (who are at substantially higher risk despite similarly low eGFR). This connection between proteinuria and kidney disease progression seems to be causal at least in part, as suggested by animal experiments and clinical observations.²⁹ It is also of note that weight reduction reduces proteinuria⁴ independent of changes in GFR.

Making good use of the data from 2 major studies, the Atherosclerosis Risk in Communities (ARIC) trial and the Cardiovascular Health Study (CHS), Elsayed et al examined an impressively large cohort of 21,258 men and women with normal serum creatinine at baseline, ie, less than 1.4 mg/dL and 1.2 mg/dL, respectively, using calibrated creatinine measurements.¹⁹ The incidence of CKD over the course of a 9.3-year observation period was defined either as an increase in serum creatinine concentration by 0.4 mg/dL or greater above baseline (with resultant serum creatinine levels greater than 1.4 mg/dL and 1.2 mg/dL, respectively), and, in a separate analysis, as a decrease of eGFR by 15 mL/min/1.73 m² or greater in patients with a baseline eGFR greater than 60 mL/min/1.73m². Because both BMI and waist-to-hip ratio values were available, the authors had the unique opportunity to compare the predictive power for renal risk of these 2 indices. A 1-SD increase in waist-to-hip ratio predicted a 22% higher risk. In contrast, 1-SD increase in BMI was not predictive of increased risk. This of course does not negate the convincing correlation between BMI and ESRD,^{8,9} but clearly documents the superiority of the waist-to-hip ratio (and presumably also waist circumference) to BMI for the detection of incipient GFR loss (unfortunately, albuminuria was not studied). This finding is of considerable public health relevance given the fact that minor reduction in kidney function is already associated with high cardiovascular risk.³⁰ The different performance of BMI and waist-to-hip ratio is not surprising: BMI fails to distinguish between weight from fat and weight from muscle or bone, and in CKD, BMI is further confounded by extracellular fluid volume expansion. In contrast, the waist-to-hip ratio (and the easier to handle and equally predictive waist circumference) captures visceral obesity, which is obviously the culprit in the genesis of progression.

Obesity is known to cause glomerular hyperfiltration, albuminuria/proteinuria, and glomerulomegaly,³¹ as well as increased mesangial matrix and mesangial cell proliferation, podocyte hypertrophy, focal segmental or global glomerulosclerosis, and interstitial fibrosis.³² But is there specific information on the relationship between visceral obesity and risk for CKD? In the Prevention of Renal and Vascular End-Stage Disease (PREVEND) study, Pinto-Sietsma³³ documented that progressively higher values of the waist-to-hip ratio were associated with a progressively higher prevalence of microalbuminuria and of diminished eGFR. This was true even in lean individuals with a BMI less than 25 kg/m². Such visceral obesity in the face of a normal BMI even led recently to the paradoxical term “normal weight obesity,”³⁴ which has been shown to be associated with high blood cholesterol and metabolic syndrome and apparently also with decreased kidney function. Several studies also noted a higher incidence of microalbuminuria in nondiabetic³⁵ or diabetic³⁶ individuals with higher waist circumference. Interestingly, in the context of the study of Foster et al,¹⁴ the PREVEND study found that high BMI tended to be associated with hyperfiltration, estimated as creatinine clearance, while central fat distribution, ie, elevated waist-to-hip ratio, was associated with hypofiltration.³³

Some insight into the pathophysiology underlying the relation between visceral obesity and kidney disease is provided by exciting new findings documenting the secretion by visceral fat of oxidized polyunsaturated fatty acids (epoxy-ketooctadecenoic acid, or EKODE) stimulating the secretion of aldosterone by the adrenal gland independent of classical secretagogues.³⁷ Aldosterone impairs podocyte function,³⁸ an effect which is reversible with eplerenone³⁹; the podocyte injury is caused by reactive oxidant species and abrogated by the radical scavenger tempol.⁴⁰ This is not the only mechanism, but space considerations do not permit discussion of further “smoking guns.”

Elsayed et al¹⁹ have finally laid to rest the once prevailing misconception that all the risk for CKD conferred by obesity is explained by diabetes and hypertension.⁴¹ One may argue that from a public health perspective, it does not matter whether the effect of obesity is an indirect

(ie, via diabetes and hypertension) or direct one—the ultimate cause is obesity, and this will be the challenge in the future. Nevertheless, Elsayed et al confirmed that the risk for CKD is found even in nondiabetic normotensive obese individuals. Although clinically manifest diabetes does not (fully) account for the risk conferred by obesity, increasing recent experimental⁴² and clinical findings⁴³ point to albuminuria and decreased kidney function in the prediabetic stage,⁴⁴ which is so common in obese patients. At least in obese primates, glomerular lesions are present even prior to the onset of overt diabetes.⁴⁵ So while diabetes is not the main culprit for the epidemiological relation between obesity and decreased kidney function, the prediabetic state of the metabolic syndrome may well be a contributing factor. Elsayed et al considered this possibility and surprisingly found that the prediction of CKD by obesity was not diminished by adjusting for many metabolic risk factors. This interesting issue deserves further study and, particularly, information from renal biopsies would be highly desirable.

If visceral obesity is such an important predictor, the question of whether waist circumference is still a valid measure of visceral fat in CKD arises, especially in view of the known catabolic state of uremic patients. Sanchez⁴⁶ addressed this issue head-on by comparing in CKD patients, mostly CKD 3 and 4, the waist circumference with the gold standard methods of computed tomography and dual-energy X-ray absorptiometry. The good news is that the simple method of measuring the waist circumference is highly correlated with visceral fat in men and women and is clearly superior to BMI, confirming the conclusion of Elsayed et al. The correlation to cardiovascular risk indicators, particularly lipids and insulin resistance, which is found in individuals without CKD, was observed in patients with CKD. The study extends previous work in patients with kidney failure treated by hemodialysis, which found good correlations between visceral fat and dyslipidemia as well as carotid atherosclerosis.^{47,48}

Finally, the paper of Chen et al⁴⁹ points to the international dimension of the problem of obesity-associated kidney disease. The authors addressed the issue of obesity-related glomerulopathy, a condition first recognized in 1974 in the United

States⁶ and subsequently well characterized in Western countries,⁷ but now slowly making its appearance in China as well. In an incredibly large 4-year sample of 10,093 renal biopsies, the authors found glomerulomegaly and proteinuria of greater than 0.4 g/d, and occasional FSGS in a low percentage (0.89%). The interesting point, however, is the finding that the frequency is on the rise. Although some of the criteria used by the authors are not directly comparable with past reports, the authors correctly conclude that obesity-associated glomerular disease is no longer a privilege of Western countries. Given the greater propensity of Asians to the sequelae of obesity,¹⁰ the magnitude of the problem may eventually even become larger in Asia.

Eberhard Ritz, MD

Klinikum der Universität Heidelberg
Heidelberg, Germany

ACKNOWLEDGEMENTS

Support: None.

Financial Disclosure: None.

REFERENCES

1. Praga M: Obesity—a neglected culprit in renal disease. *Nephrol Dial Transplant* 17:1157-1159, 2002
2. Bonnet F, Deprele C, Sassolas A, et al: Excessive body weight as a new independent risk factor for clinical and pathological progression in primary IgA nephritis. *Am J Kidney Dis* 37:720-727, 2001
3. Morales E, Valero MA, Leon M, et al: Beneficial effects of weight loss in overweight patients with chronic proteinuric nephropathies. *Am J Kidney Dis* 41:319-327, 2003
4. Bello AK, de Zeeuw D, El Nahas M, et al: Impact of weight change on albuminuria in the general population. *Nephrol Dial Transplant* 22:1619-1627, 2007
5. Chagnac AWT, Korzets A, Ramadan E, Hirsch J, Gafter U: Glomerular hemodynamics in severe obesity. *Am J Physiol Renal Physiol* 278:F817-F822, 2000
6. Weisinger JR, Kempson RL, Eldridge FL, Swenson RS: The nephrotic syndrome: a complication of massive obesity. *Ann Intern Med* 81:440-447, 1974
7. Kambham N, Markowitz GS, Valeri AM, et al: Obesity-related glomerulopathy: an emerging epidemic. *Kidney Int* 59:1498-1509, 2001
8. Iseki K, Ikemiya Y, Kinjo K, et al: Body mass index and the risk of development of end-stage renal disease in a screened cohort. *Kidney Int* 65:1870-1876, 2004
9. Hsu CY, McCulloch CE, Iribarren C, et al: Body mass index and risk for end-stage renal disease. *Ann Intern Med* 144:21-28, 2006

10. WHO Expert Consultation: Appropriate body-mass index for Asian populations and its implications for policy and intervention strategies. *Lancet* 363:157-163, 2004
11. Ejerblad E, Forel CM, Lindblad P, et al: Obesity and risk for chronic renal failure. *J Am Soc Nephrol* 17:1695-1702, 2006
12. Brenner BM, Garcia DL, Anderson S: Glomeruli and blood pressure. Less of one, more the other? *Am J Hypertens* 1:335-347, 1988
13. Barker DJ, Osmond C, Forsen TJ, et al: Trajectories of growth among children who have coronary events as adults. *N Engl J Med* 353:1802-1809, 2005
14. Foster MC, Hwang S-J, Larson MG, et al: Overweight, obesity, and the development of stage 3 CKD: The Framingham Heart Study. *Am J Kidney Dis* 52:39-48, 2008
15. Poggio ED, Rule AD: Can we do better than a single estimated GFR threshold when screening for chronic kidney disease? *Kidney Int* 72:534-536, 2007
16. Calabro P, Yeh ET: Intra-abdominal adiposity, inflammation, and cardiovascular risk: new insight into global cardiometabolic risk. *Curr Hypertens Rep* 10:32-38, 2008
17. Ritz E: Metabolic syndrome and kidney disease. *Blood Purif* 26:59-62, 2008
18. Klein S, Fontana L, Young VL, Coggan AR, et al: Absence of an effect of liposuction on insulin action and risk factors for coronary heart disease. *N Engl J Med* 350:2549-2557, 2004
19. Elsayed EF, Sarnak MJ, Tighiouart H, et al: Waist-to-hip ratio, body mass index, and subsequent kidney disease and death. *Am J Kidney Dis* 52:29-38, 2008
20. Smith HW: *The Kidney—Structure and Function in Health and Disease*. Oxford, UK, Oxford University Press, 1964, p 543
21. Bosma RJ, van der Heide JJ, Oosterop EJ, et al: Body mass index is associated with altered renal hemodynamics in non-obese healthy subjects. *Kidney Int* 65:259-265, 2004
22. Tomaszewski M, Charchar FJ, Maric C, et al: Glomerular hyperfiltration: a new marker of metabolic risk. *Kidney Int* 71:816-821, 2007
23. Krikken JA, Lely AT, Bakker SJ, Navis G: The effect of a shift in sodium intake on renal hemodynamics is determined by body mass index in healthy young men. *Kidney Int* 71:260-265, 2007
24. Schoner W, Scheiner-Bobis G: Endogenous and exogenous cardiac glycosides: their roles in hypertension, salt metabolism, and cell growth. *Am J Physiol Cell Physiol* 293:C509-C536, 2007
25. Hall JE: The kidney, hypertension, and obesity. *Hypertension* 41:625-633, 2003
26. Dengel DR, Goldberg AP, Mayuga RS, et al: Insulin resistance, elevated glomerular filtration fraction, and renal injury. *Hypertension* 28:127-132, 1996
27. Hall JE, Brands MW, Mizelle HL, et al: Chronic intrarenal hyperinsulinemia does not cause hypertension. *Am J Physiol* 260:F663-F669, 1991
28. D'Agati VD, Markowitz GS: Supersized kidneys: Lessons from the preclinical obese kidney. *Kidney Int* 73:909-910, 2008
29. Remuzzi G, Bertani T. Pathophysiology of progressive nephropathies. *N Engl J Med* 339:1448-1456, 1998
30. Keith DS, Nichols GA, Gullion CM, et al: Longitudinal follow-up and outcomes among a population with chronic kidney disease in a large managed care organization. *Arch Intern Med* 164:659-663, 2004
31. Lastra G, Manrique C, Sowers JR: Obesity, cardio-metabolic syndrome, and chronic kidney disease: the weight of the evidence. *Adv Chronic Kidney Dis* 13:365-373, 2006
32. Serra A, Romero R, Lopez D, Navarro M, et al: Renal injury in the extremely obese patients with normal renal function. *Kidney Int* 73:947-955, 2008
33. Pinto-Sietsma SJ, Navis G, Janssen WM, et al: A central body fat distribution is related to renal function impairment, even in lean subjects. *Am J Kidney Dis* 41:733-741, 2003
34. Romero-Corral A, Lopez-Jimenez F, Boarin S, Sierra-Johnson J, Somers VK: Normal weight obesity: A risk factor for cardiometabolic dysregulations. *J Am Coll Cardiol* 51:A350, 2008 (abstr 814-4, suppl 1)
35. Bonnet F, Marre M, Halimi JM, et al: Waist circumference and the metabolic syndrome predict the development of elevated albuminuria in non-diabetic subjects: the DESIR Study. *J Hypertens* 24:1157-1163, 2006
36. de Boer IH, Steffes MW: Glomerular filtration rate and albuminuria: twin manifestations of nephropathy in diabetes. *J Am Soc Nephrol* 18:1036-1037, 2007
37. Goodfriend TL, Ball DL, Egan BM, Campbell WB, Nithipatikorn K: Epoxy-keto derivative of linoleic acid stimulates aldosterone secretion. *Hypertension* 43:358-374, 2004
38. Nagase M, Yoshida S, Shibata S, et al: Enhanced aldosterone signaling in the early nephropathy of rats with metabolic syndrome: possible contribution of fat-derived factors. *J Am Soc Nephrol* 17:3438-3446, 2006
39. Nagase M, Shibata S, Yoshida S, et al: Podocyte injury underlies the glomerulopathy of Dahl salt-hypertensive rats and is reversed by aldosterone blocker. *Hypertension* 47:1084-1093, 2006
40. Shibata S, Nagase M, Yoshida S, et al: Podocyte as the target for aldosterone: roles of oxidative stress and Sgk1. *Hypertension* 49:355-364, 2007
41. Bakker SJ, Gansevoort RT, de Zeeuw D: Metabolic syndrome: a fata morgana? *Nephrol Dial Transplant* 22:15-20, 2007
42. Nagai Y, Yao L, Kobori H, et al: Temporary angiotensin II blockade at the prediabetic stage attenuates the development of renal injury in type 2 diabetic rats. *J Am Soc Nephrol* 16:574-575, 2005
43. Mykkanen L, Zaccaro DJ, Wagenknecht LE, et al: Microalbuminuria is associated with insulin resistance in nondiabetic subjects: the insulin resistance atherosclerosis study. *Diabetes* 47:793-800, 1998
44. Brantsma AH, Bakker SJ, Hillege HL, et al: Urinary albumin excretion and its relation with C-reactive protein and the metabolic syndrome in the prediction of type 2 diabetes. *Diabetes Care* 28:2525-2530, 2005
45. Caramori ML, Hansen BC, Mauer M, et al: Renal structural-functional relationships in spontaneously obese rhesus monkeys (RhM). *Renal Week 2007: American Society of Nephrology Annual Meeting, San Francisco, CA, October 31 to November 5, 2007 (abstr F-PO331)*

46. Sanches FMR, Avesani CM, Kamimura MA, et al: Waist circumference and visceral fat in CKD: A cross-sectional study. *Am J Kidney Dis* 52:66-73, 2008
47. Odamaki M, Furuya R, Ohkawa S, et al: Altered abdominal fat distribution and its association with the serum lipid profile in non-diabetic haemodialysis patients. *Nephrol Dial Transplant* 14:2427-2432, 1999
48. Yamauchi T, Kuno T, Takada H, et al: The impact of visceral fat on multiple risk factors and carotid atherosclerosis in chronic haemodialysis patients. *Nephrol Dial Transplant* 18:1842-1847, 2003
49. Chen H-M, Li S-J, Chen H-P, Wang Q-W, Li L-S, Liu Z-H: Obesity-related glomerulopathy in China: A case series of 90 patients. *Am J Kidney Dis* 52:58-65, 2008